

Stephen Purdue

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EDUCATION

Royal Holloway, University of London

Master of Science - Artificial Intelligence

London, UK

2026 - Expected 2027

University of Winchester

Bachelor of Science - Computer Game Development

Winchester, UK

2023 - 2026

EXPERIENCE

Sous-Chef, Chef de Partie

Mustang Group, Southampton, UK

2021 - Present

- Led a kitchen of five through high-volume, deadline-driven service under constant time pressure, requiring rapid problem-solving, attention to detail, and direct communication with other colleagues in an agile environment.
- Mentored and trained 4-8 junior chefs and new starters per intake, maintaining consistent service standards and reducing onboarding time during high staff turnover.
- Maintained stock rotation and portion-control procedures to reduce food waste and control ingredient costs across a high-volume kitchen.

PROJECTS

Stock Market Prediction Engine

[GitHub](#)

- Developed an LSTM model in PyTorch to forecast daily stock prices from historical time-series analysis data, reducing prediction error by 81% versus a moving-average baseline. Containerised the full stack with Docker Compose, reducing local setup time to under 5 minutes.

Adaptive Combat Simulator

[GitHub](#)

- Developed an adaptive combat simulator, achieving a 75% win rate against a baseline agent, with overall reward improving from 32% to 87%, measured over 30 unique evaluation simulations by training a PPO-agent using Python and TensorFlow.

Procedural Generation Algorithm

[GitHub](#)

- Engineered a scalable procedural generation system in Unity using C# and Binary Space Partitioning to recursively subdivide space into room layouts and connective corridors in linear time. Reduced level build-time by 77% by implementing modular, parameter-driven logic.

Ad Trigger Discovery Engine (Cursor Hackathon)

[GitHub](#)

- Individually built the analysis backend for an ad-tech pipeline, generating synthetic user prompts across intent buckets and running BM25/embedding-based retrieval to surface and rank high-intent keyword triggers from collected brand data, with full source traceability for review.

TECHNICAL SKILLS

Languages: Python, C#, C++, SQL, Bash

Technologies: Linux, Git, Docker, AWS (S3, SageMaker, Bedrock), CI/CD, PyTorch, TensorFlow

INTERESTS

- Snowboarding, Competitive Gaming